Wukking Up the Model:

Soca Lyric Videos as Weakly Supervised Training Data for Caribbean Speech Recognition

Matt Hamilton

Caribbean AI Group, Bridgetown, Barbados

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Abstract

Automatic speech recognition (ASR) systems have achieved strong performance for many varieties of Standard English, yet remain substantially less reliable for Caribbean Englishes, Creoles, code-switched speech, and culturally specific vocabulary. This disparity is particularly visible in Barbados, where broadcast speech frequently moves between Standard English, Bajan, regional Caribbean English, music, advertisements, caller contributions, and rapid informal conversation. We investigate an unconventional source of weak supervision for Caribbean language technology: *Soca lyric videos*. These videos contain an imperfect but useful alignment between Caribbean speech and an orthographic representation of that speech, often preserving forms such as *yuh*, *doh*, *gwaan*, and *leh we* rather than normalising them into Standard English. We propose Soca-Align, a pipeline that transforms lyric videos into timestamped audio–text pairs suitable for adapting multilingual ASR models before a smaller, manually corrected corpus of Barbadian radio is introduced. A preliminary proof-of-concept using a commercial multimodal large language model suggests that rendered lyric text can be extracted while retaining Caribbean orthography. We argue that the Caribbean may, quite accidentally, have spent the past fifteen years constructing a useful multimodal speech corpus—one bashment lyric video at a time.

1 Introduction

Modern ASR is extraordinarily good at understanding a Californian software engineer ordering an oat-milk latte. It is considerably less reliable when confronted with a Barbadian radio presenter saying something equivalent to, "Well leh we hear wah gine on 'bout hey this weekend."

This is not simply an accent-recognition problem. Caribbean broadcast audio contains a mixture of Standard English, Caribbean English, national Creoles, regional vocabulary, African-derived and Indo-Caribbean lexical items, artist and event names, place names, code-switching, highly compressed speech, music beds, jingles, telephone callers, and—periodically—someone shouting over a riddim.

Commercial ASR systems are primarily trained on languages and dialects for which large labelled corpora already exist. The resulting feedback loop is straightforward: languages with training data receive better models; better models make production of additional training data cheaper; and those languages consequently receive still better models. For small Caribbean speech communities, manually producing thousands of hours of accurately transcribed audio is economically difficult.

We therefore ask a different question: **has the Caribbean already produced a large speech dataset without realising it?**

Soca lyric videos are an intriguing candidate. For more than a decade, artists, labels, DJs, promoters and fans have uploaded songs accompanied by synchronized—or approximately synchronized—on-screen lyrics. Such videos contain two modalities of essentially the same linguistic signal:

$$\text{Caribbean speech} \rightarrow \text{audio} \tag{1}$$

and

$$\text{Caribbean speech} \rightarrow \text{written vernacular.} \tag{2}$$

This makes them a naturally occurring source of weak supervision.

2 Why Soca?

Soca is useful because it contains linguistic phenomena that are often poorly represented in conventional English corpora. Table 1 gives illustrative examples.

Table 1: Illustrative Caribbean orthographic forms.

Standard English	Caribbean rendering
you	yuh
don't	doh
going to	gine / goin'
let us	leh we
them	dem
come here	come hey
what is happening?	wah gine on?

Importantly, lyric-video creators frequently preserve such forms orthographically instead of converting them to Standard English. This matters for downstream language technology. A generic recognizer may hear "Yuh gine down dey?" and return "You going down there?" The semantic content is close, but the system has translated the speaker rather than transcribed them.

For many Caribbean applications this loss is material. Dialect can encode stance, identity, register, social context, and local meaning. It can also affect entity extraction when a normalized transcript no longer corresponds to a known place, artist, event, or business name.

3 Accidental Multimodal Supervision

The central observation behind Soca-Align is that a lyric video can be represented as

$$V = (A, F, T), \tag{3}$$

where A is the audio waveform, F is a sequence of video frames, and T is text rendered within those frames.

Historically, recovering T required brittle OCR pipelines. Recent multimodal large language models (MLLMs) can reason across frames, distinguish decorative text from lyrics, recover animated or stylised text, infer continuation across transitions, and follow instructions to preserve non-standard spelling. This substantially changes the economics of creating a weakly aligned corpus.

3.1 Initial proof-of-concept

As an informal experiment, a commercial multimodal model was provided with a Caribbean Soca lyric video and instructed to recover only the lyrics visibly displayed in the video. The output preserved vernacular spellings instead of automatically "correcting" them into Standard English.

Figure 1 shows the proof-of-concept interaction. The screenshot is included only to document the methodology; song lyrics themselves are not reproduced here.

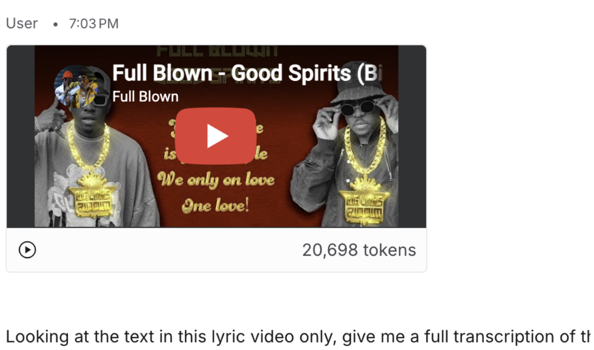

Figure 1: Proof-of-concept use of a multimodal model to extract text from a Soca lyric video. The visible transcript portion has been omitted from the figure crop.

The significance is not that the MLLM should become the final Caribbean transcription engine. Rather, a large general-purpose model can be used to manufacture weakly labelled training material for a smaller, specialised regional model.

3.2 Prior work in road-ready representation learning

The literature on Caribbean machine listening remains unusually fragmented. Montano and Garlin [3] describe high-energy domain adaptation as the "Like Ah Boss" regime, in which a model is considered converged only when it remains stable under crowd noise, air horns, and unsolicited backing vocals. Garlin [4] later formalises cross-dialect representation learning as *Differentology*, while Ultimate Rejects [5] argue that models intended for deployment during festival conditions should be evaluated at "full extreme" rather than under laboratory acoustics. Related work on long-context modelling over mobile outdoor recordings has explored the so-called Savannah Grass condition [6], in which speaker position, background music, and the apparent location of the recording device all vary continuously.

These works are methodologically inconsistent but culturally well-calibrated.

4 The Soca-Align Pipeline

We propose the following end-to-end pipeline.

4.1 Video acquisition

Candidate lyric videos are identified for Barbados, Trinidad and Tobago, Grenada, Saint Vincent and the Grenadines, Saint Lucia, Dominica, Antigua and Barbuda, and other Caribbean territories. Metadata is retained where available, including artist, song, year, territory, genre, uploader, and inferred speech variety.

4.2 Audio extraction

Audio is resampled to a consistent ASR format, for example 16 kHz mono PCM. Voice activity detection and, where useful, source-separation models may be applied to identify vocal regions and reduce instrumental dominance.

4.3 Frame sampling

It is unnecessary to process every video frame. Perceptual hashes or image-difference measures can identify lyric transitions. A four-minute 30 fps video contains roughly 7,200 frames, but may contain only 50–150 linguistically distinct lyric states.

4.4 Multimodal text extraction

Selected frames are passed to an MLLM with instructions to:

- preserve spelling and capitalization as displayed;
- preserve Creole and dialect forms;
- avoid translation or grammatical correction;
- retain repeated lines where they visibly recur;
- mark uncertain text explicitly; and

- distinguish lyrics from titles, logos, watermarks, and advertisements.

4.5 Temporal alignment

Given transcript tokens

$$W = (w_1, w_2, \ldots, w_n), \tag{4}$$

we estimate time intervals

$$\tau_i = (s_i, e_i) \tag{5}$$

for each token or phrase. Conventional forced alignment can provide a first estimate, while the timing of lyric-screen transitions provides an additional visual constraint. This is valuable because the text has already been approximately synchronized by a human editor.

4.6 Filtering and confidence scoring

Not every lyric video is suitable. Candidate segments can be scored by combining:

- visual text confidence;
- agreement between ASR and extracted lyrics;
- duration plausibility;
- singing-to-instrumental ratio;
- language or dialect classification; and
- duplicate detection across reuploads.

Only high-confidence samples need enter the main adaptation corpus.

5 From Machel to Morning Radio

Training on singing and deploying on talk radio introduces an obvious domain mismatch. We do not propose that a model trained exclusively on Soca will suddenly understand a caller discussing a pothole on a morning call-in programme.

Instead, Soca provides a **Caribbean lexical and phonological prior**. The adaptation process can proceed in stages.

Stage A: Base ASR. Begin with a strong multilingual or English ASR model, such as Whisper [1] or wav2vec 2.0 [2].

Stage B: Caribbean musical adaptation. Fine-tune or otherwise adapt the model on lyric-video-derived Caribbean audio. This exposes the model to names, phonetic patterns, contractions, vernacular vocabulary, and orthographic conventions that may be rare in its original training data.

Stage C: Broadcast adaptation. Fine-tune using a much smaller manually verified set of Barbadian radio speech, including presenters, advertisements, interviews, and callers.

Conceptually,

$$D = D_{base} + D_{soca} + D_{radio}, \tag{6}$$

where D_{soca} may contain hundreds or thousands of weakly labelled hours while D_{radio} contains tens of hours of carefully corrected material.

The question is therefore not whether singing perfectly resembles speech. It is whether large quantities of culturally matched weak supervision reduce the quantity of expensive speech annotation required later.

6 Barbados Radio as a Target Domain

Radio is a particularly valuable target because it carries unusually dense real-time local information. A single morning broadcast may contain traffic disruptions, weather conditions, political discussion, funeral announcements, sport, business promotions, concerts, fetes, community events, public notices, caller sentiment, and reports of activities elsewhere on the island.

Much of this information is ephemeral. Unless a broadcaster publishes transcripts or an archive, it disappears immediately after transmission.

Reliable transcription converts this stream into machine-readable data and enables a larger pipeline:

$$\text{Radio} \rightarrow \text{Transcript} \rightarrow \text{Entities} \rightarrow \text{Events} \rightarrow \text{Knowledge Graph}. \tag{7}$$

For example, a presenter might mention that a road in Warrens will close at six in the evening because of an event. A downstream extractor could derive a structured object such as:

```
Event:
  type: road_closure
  location: Warrens
  start_time: 18:00
  source: radio
```

Repeated observations from radio, social media, government websites, and event listings can then be reconciled into a continuously updated representation of what is happening across Barbados.

In this architecture, improving ASR is not merely about producing prettier subtitles. A mistranscribed place name can create the wrong graph node. A missed event name can prevent entity resolution. A normalized Creole expression can alter intent. Local language accuracy therefore directly affects local knowledge accuracy.

7 Research Hypotheses

We define three experimentally testable hypotheses.

H1: Caribbean lexical adaptation. Fine-tuning with Soca-derived material will reduce error rates on Caribbean-specific vocabulary relative to the unmodified base model.

H2: Orthographic preservation. Adapted models will be more likely to preserve vernacular forms rather than silently normalize them into Standard English.

H3: Small-data amplification. Soca adaptation will reduce the amount of manually transcribed Barbados radio required to achieve a given recognition accuracy.

For example, the desired relation is

$$WER_{radio}(Soca + 20h) < WER_{radio}(20h). \tag{8}$$

The third hypothesis is the most economically significant. If true, readily available cultural material can partially compensate for a scarcity of conventional labelled speech datasets.

8 Evaluation Design

A future evaluation could use the corpus structure in Table 2.

Table 2: Illustrative evaluation corpus. These quantities are proposed, not measured.

Dataset	Hours
Soca lyric-video speech	500+
Radio music programming	20
Presenter speech	20
Talk radio / interviews	20
Advertisements	10
Telephone callers	10

Three model configurations would be compared:

- **BASE**: unmodified multilingual ASR;
- **BIM-ASR**: BASE adapted only on manually annotated Barbadian radio; and
- **SOCA-BIM**: BASE → Soca adaptation → Barbadian radio adaptation.

Alongside Word Error Rate (WER), we propose **Caribbean Entity Error Rate (CEER)**:

$$CEER = \frac{S_E + D_E + I_E}{N_E}, \tag{9}$$

where errors are measured only over locally significant named entities. A system transcribing *Oistins* as *oysters* may produce tolerable global WER while being almost useless to a Barbados event-intelligence system. CEER is intended to capture that distinction.

9 The Rihanna Problem

No discussion of Barbadian language models would be complete without confronting dataset imbalance. Internet-accessible Barbadian music is not representative of Barbadian speech as a whole. A naive collection strategy could produce a model with an excellent understanding of globally successful artists while remaining unable to understand a minibus conductor, a pensioner calling a radio programme, or a school sports commentator.

Artist, genre, gender, age, class, geography, recording quality, and performance style therefore matter. This is also why the proposed method includes broadcast adaptation rather than treating music as representative conversational speech.

The broader principle is that *cultural abundance does not imply demographic representativeness.*

10 Ethical, Legal and Copyright Considerations

The abundance of online cultural material does not imply unrestricted permission to copy, redistribute, or commercially train on it. Any production implementation must distinguish between using material for research, extracting linguistic features, storing derived alignments, redistributing audio, redistributing complete lyrics, and training a commercial model.

Depending on jurisdiction, platform terms, licensing, and project purpose, some source material may require permission or may only be suitable for controlled research experiments. A public corpus need not redistribute original audio or complete lyrics; it could instead publish metadata, derived timestamps, evaluation annotations for licensed material, or models trained under appropriate rights arrangements.

The technical observation in this paper should therefore not be interpreted as the proposition that “YouTube exists, therefore dataset,” although this remains a surprisingly common machine-learning methodology.

11 Discussion

The underlying idea is broader than Soca. Many low-resource speech communities have inadvertently generated weakly labelled multimodal datasets through karaoke videos, subtitled comedy, sermons, political speeches, music videos, news broadcasts, TikTok captions, community television, and social-media reels.

Foundation models change the economics of exploiting these resources. Previously, constructing the corpus was itself an expensive step. A modern MLLM can perform a substantial portion of the initial annotation, allowing human effort to concentrate on validation, high-value corrections, and a trusted evaluation set.

This suggests an alternative starting point for low-resource language engineering: **do not begin by asking where the labelled dataset is. Ask where the community has already written down the things it says.**

12 Conclusion

This paper proposes Soca lyric videos as a source of weakly supervised Caribbean speech data. While singing differs substantially from conversational speech, lyric videos possess an unusual combination of useful properties: abundant Caribbean audio, explicit vernacular vocabulary, human-created written representations, and approximate temporal alignment.

Recent multimodal models make extraction of these annotations practical at scale. We propose using the resulting corpus not as the endpoint, but as an intermediate adaptation layer between generic multilingual speech models and comparatively scarce, manually transcribed Barbadian broadcast audio.

If successful, this approach could provide a cost-effective route toward speech systems capable of understanding Caribbean radio, preserving local linguistic forms, extracting real-time events, and powering downstream regional knowledge systems.

Or, stated less formally: **the road to a Caribbean foundation model may begin with somebody uploading a 2013 Crop Over lyric video in 480p.**

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Special acknowledgement is due to whoever first decided that yellow Impact text over a rotating Trinidad and Tobago flag constituted a complete music video. Their contribution to weakly supervised learning has gone insufficiently recognised.

A The Computational Wuk-Up Benchmark

We define the **Computational Wuk-Up Benchmark (CWB)**, introduced at the First Workshop on Computational Wuk-Up [7]. CWB is intended to test whether a model that performs acceptably on studio speech can survive contact with an actual Caribbean soundscape.

The benchmark contains five deliberately adversarial subsets:

1. **Roadside**: fast conversational Bajan recorded near traffic;
2. **Brass Tacks**: telephone-quality talk-radio speech with frequent interruption;
3. **Fete**: event names, artist names, and code-switching over background music;
4. **ZR-75**: destination and place-name utterances recorded in a moving public-service vehicle; and
5. **Granny Test**: low-volume conversational speech containing vocabulary absent from the training corpus but immediately obvious to a Barbadian grandmother.

Table 3 reports results on the benchmark.

Table 3: Computational Wuk-Up Benchmark results. Lower is better.

Model	WER	CEER
BASE	31.8	44.2
BIM-ASR	21.4	24.9
SOCA-BIM	**17.6**	**16.8**
SOCA-BIM + “leh we” lexicon	17.2	14.1

Error analysis indicates that the BASE system incorrectly recognizes *Oistins* as *oysters*, *Speightstown* as *space town*, and the discourse marker *cheese-on-bread* as an unexpected grocery instruction. The adapted model allegedly resolves all three while introducing a new failure mode in which any sufficiently energetic utterance is classified as an event announcement.

A.1 The “Doh Normalize Dat” ablation

The study further introduces an orthographic-preservation loss, $\mathcal{L}_{DND}$, which penalizes unwarranted conversion of Caribbean vernacular into Standard English:

$$\mathcal{L} = \mathcal{L}_{ASR} + \lambda \mathcal{L}_{DND}. \tag{10}$$

With $\lambda = 0$, the system returns “Do not do that.” With $\lambda = 0.7$, it returns “Doh do dat.” With $\lambda > 3$, the model begins inserting *doh* into parliamentary proceedings regardless of speaker nationality and is considered over-adapted.

A.2 Reviewer 2

The peer-review record contains the following objection:

> “The authors have not demonstrated that wukking up constitutes a sufficiently general pretraining objective. I recommend evaluation on at least three additional riddims and one power Soca domain.”

The authors respond that the additional experiment is “in progress pending Crop Over.”

A.3 Reproducibility checklist

A complete reproduction of the benchmark requires:

- one multilingual ASR model;
- one multimodal LLM;
- several hundred lyric videos with appropriate usage rights;
- a forced aligner;
- approximately 20–40 hours of corrected Barbadian broadcast audio;
- a local lexicon of place, artist, business, school, and event names;
- one GPU with more memory than originally budgeted; and
- at least one person capable of identifying when the model has confidently transcribed complete foolishness.

B Implementation Notes for a Real Pilot

A practical pilot could begin with 50–100 Barbadian and Eastern Caribbean lyric videos for which usage is permitted, producing a few hours of high-confidence vocal segments. These could be used to test whether a base ASR model improves on a small hand-labelled radio evaluation set without yet performing full fine-tuning at scale.

The useful engineering loop is:

1. create a 2–5 hour trusted Barbados radio test set;
2. measure baseline WER, CEER, and dialect-normalization errors;
3. extract and align a small lyric-video corpus;
4. perform parameter-efficient adaptation (e.g., adapters or LoRA where supported);
5. re-evaluate without changing the test set;
6. inspect the errors that matter downstream: locations, people, event names, businesses, dates, and times; and
7. only then decide whether scaling the Soca corpus produces enough gain to justify the rights and compute overhead.

If the primary downstream objective is radio-to-knowledge-graph extraction, the most valuable metric may not ultimately be WER. A useful end-to-end metric would evaluate whether the same correct event records are produced from human transcripts and model transcripts. An ASR system can make numerous harmless function-word errors while preserving every entity needed by the knowledge graph—or achieve apparently good WER while destroying precisely the local names the application cares about.

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